

20ME32006 – Refrigeration and Air Conditioning **(Professional Elective-I)**

B.Tech. ME - III Year II Semester

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Prerequisites: 20ME21004 - Thermodynamics
 20ME22001 - Thermal Engineering

Note: Refrigeration And Air-Conditioning Data Book by Anand V. Domkundwar and V.M. Domkundwar is permitted in examination

Course Objectives: This Course covers:

1. Basics of Refrigeration
2. Vapor compression Refrigeration (VCR)
3. Refrigerants used in VCR
4. Non-Conventional Refrigeration systems
5. Cooling/ heating load calculations for a given application

Course Outcomes: At the end of the course, the student would be able to:

- CO1: Analyze and find the necessity of Refrigeration
- CO2: Comprehend Vapour compression refrigeration cycles through P-V and T-S diagrams
- CO3: Select refrigerants and components applied to VCR
- CO4: Outline alternate refrigeration systems to VCR
- CO5: Design the Air Conditioning System for human comfort

UNIT-I:

Introduction: Necessity and applications, Unit of refrigeration, Mechanical Refrigeration systems

Air Refrigeration Cycles: Bell Coleman cycle and Brayton cycle; Open and Dense air refrigeration systems, Actual air refrigeration system, Refrigeration needs for Air crafts

UNIT-II:

Vapour Compression Refrigeration (VCR): Working principle and essential components for VCR plant, Simple Vapour compression refrigeration cycle, Coefficient of Performance (COP), Representation of cycle with T-S and P-H charts, Effect of sub cooling and super heating, Cycle analysis, Actual cycle, Influence of various parameters on system performance

UNIT-III:

Components of VCR system: Classification and Working Principle of Compressors, Condensers, Expansion devices, Evaporators

Refrigerants: Desirable properties, common refrigerants, Nomenclature, Ozone Depletion, Global Warming, Azeotropes and Zeotropes, General classification, comparison, Advantages and Disadvantages; Introduction to Nano refrigerants

UNIT-IV:

Vapour Absorption Refrigeration: Calculation of maximum COP, Description and working principle of NH₃, Water systems, Li – Br system; Principle and operation of Three Fluid absorption system, salient features, Steam Jet Refrigeration System – Working Principle and Basic Components

Non-Conventional Refrigeration Systems: Principle and operations of (i) Thermoelectric refrigerator (ii) Vortex tube or Hilsch tube

UNIT-V:

Air Conditioning: Psychometric Properties and Processes - Sensible and latent heat loads, Characterization, Need for Ventilation, Consideration of Infiltration, Load concepts of RSFH, ASHF, ESHF and ADP; Concept of human comfort and effective temperature –Comfort Air conditioning, Industrial air conditioning and their requirements, Load Calculations; Air Conditioning systems - Classification of equipment based on cooling, heating, humidification and dehumidification

TEXT BOOKS:

1. Refrigeration and Air Conditioning, CP Arora, TMH, 2008
2. A course in Refrigeration and Air- Conditioning, Domkundwar, Arora, Dhanpat Rai and Co, 2018

REFERENCE BOOKS:

1. Refrigeration & Air conditioning, W. Stoecker, McGraw Hill Higher education, 1989
2. Principles of Refrigeration, Roy. J. Dossat , PEARSON publications, 1989
3. Carrier, Hand Book of Air conditioning system design –McGraw Hill, 2009
4. Refrigeration & Air conditioning, F.Stoecker & Jerold. W. Jones, McGraw Hill, 1982
5. Refrigeration And Air-Conditioning - Data Book, Anand V. Domkundwar V.M. Domkundwar, Dhanpat Rai and co publishers